

README

BOB-207 — QA evidence: precondition/repair agreement on the ownership property

Revision: 1 **Last modified:** 2026-08-27T00:00:00Z **Item:** BOB-207 ·
Feature: 002-user-owned-downloads **Artifacts under test:** scripts/
ownership_repair.sh, scripts/lib/ownership.sh, tests/unit/
test_ownership_repair.sh, tests/unit/
test_ownership_gid_agreement.sh **Baseline commit:**
a3e1141777914f3d71d62a6cd32a4c79c7f43842 (HEAD at capture time)

Why this directory exists

An earlier report of this work claimed the pre-re-base repair suite was “146 passed / 0 failed — completely blind” to a mutation. **That figure does not reproduce from committed history and is WITHDRAWN.** It was measured against the *uncommitted working-tree* state of tests/unit/test_ownership_repair.sh (which was already M before this work began), not against HEAD. That intermediate was overwritten by the re-base and not preserved, so the number cannot be re-derived — the defect is the missing snapshot, not the measurement.

Every figure below is re-derived from committed history in this session, with the sha256 of the exact artifact each run used. This is the third chain to lose an uncommitted intermediate (after BOB-195 and BOB-102); the remedy adopted by both — snapshot before editing — is adopted here.

The corrected claim

The committed suite **does** catch a selects-nothing walk. What it cannot see is the loss of the **namespace fallback** — `unshare_chown_paths()`, the only code path able to repair the production defect (a foreign uid, which a plain `chown` cannot touch: `EPERM`). The old fixture used a *gid* mismatch, which a plain `chgrp` repairs, so that path was never reached and its removal was invisible.

#	Suite	Repair artifact	Mutation	Result	Verdict
A	committed	committed	none (control)	86 / 0 / 0 · rc 0	baseline
B	committed	committed	<code>unshare_chown_paths</code> disabled	86 / 0 / 0 · rc 0	BLIND
C	committed	committed	walk selects nothing (-false)	61 / 18 / 1 · rc 1	catches
D	re-based	narrowed	none (control)	146 / 0 / 0 · rc 0	baseline
E	re-based	narrowed	<code>unshare_chown_paths</code> disabled	101 / 37 / 1 · rc 1	CATCHES
F	re-based	narrowed	walk selects nothing (-false)	116 / 23 / 1 · rc 1	catches

B vs E is the load-bearing pair. Identical mutation, opposite verdicts: the committed suite ships green while the production repair path is gone; the re-based suite fails 37 checks. That is what the re-base bought, and it is a narrower, stronger claim than the withdrawn one.

C is recorded because it *refutes* the withdrawn “completely blind” wording: on the selects-nothing axis the old suite was never blind.

Artifact provenance (sha256, first 16)

Config	suite	repair
A	e3aaa7ed49e663f0	ae025b74602414ca

Config	suite	repair
B	e3aaa7ed49e663f0	d55a0c098129436e
C	e3aaa7ed49e663f0	8b8b0b7b91a095b0
D	03fb989d2781acdd	47889d4cd25fcdb1
E	03fb989d2781acdd	0c470e9aeb5a58bf
F	03fb989d2781acdd	9874e64eea4b4e6f

A/B/C share the committed suite; D/E/F share the re-based suite. The repair hash differs per mutation, so no two rows can be the same run mislabelled.

Commands

```
# committed-history tree (configs A-C)
git show HEAD:scripts/ownership_repair.sh > $T/scripts/
ownership_repair.sh
git show HEAD:scripts/lib/ownership.sh > $T/scripts/
lib/ownership.sh
git show HEAD:tests/unit/test_ownership_repair.sh > $T/tests/unit/
test_ownership_repair.sh

# mutation B/E – disable the fallback at its source (both call
sites: the
# batch path at flush_batch and the per-path path in repair_one)
# UNSHARE_ERR="$( "${RUNTIME}" unshare chown -h -- 0:0 "$@" 2>&1
>/dev/null)" && return 0
# return 1
# becomes
# UNSHARE_ERR="MUTATION: namespace fallback disabled"
# return 1

# mutation C/F – the walk selects nothing
# find_args+=(-false -printf '%U\t%G\t%m\t%p\0')

timeout 900 nice -n 19 ionice -c 3 bash $T/tests/unit/
test_ownership_repair.sh
```

Raw output for every configuration is in runs/.

Final state of the shipped tree

Suite	Result
tests/unit/ test_ownership_gid_agreement.sh	5 / 0 / 0 · rc 0 (runs/ agreement_suite_final.txt)

Suite	Result
tests/unit/test_ownership_repair.sh	146 / 0 / 0 · rc 0 (runs/ config_D.txt)
tests/pre_build/ test_check_cm_ownership_invariants.sh	42 / 42 · rc 0 (runs/ pre_build_gate_final.txt)

Reverting the one-line predicate change restores the original BOB-207 failure verbatim — runs/mutation_revert_predicate.txt, 4 passed / 1 failed / rc 1.

A mutation that did NOT hold, recorded

Removing only `|| unshare_chown_paths` from `flush_batch` left the suite green. Not a fake-pass: `repair_one()` is a second call site, so the tool legitimately still repaired. Mutations B/E therefore disable the function at its source. Recorded because a mutation that fails to fail is a result, not a nuisance.

Fixture note

The re-based suite seeds a **real foreign uid** via `podman unshare chown 1:1` (host-visible 100000), replacing the gid proxy. Files and symlinks carry the foreign uid; directories stay operator-owned except Case 22's declared root — see the DECLARED GAPS output in `runs/config_D.txt` and `FOLLOWUP_ITEMS.md` ITEM 1.